Telecom NetOps AI Assistant

Autonomous 5G Incident Retrieval, Multi-Agent Diagnosis, and Decision Support

5G | Smarter Networks | Faster Diagnosis | Reliable Operations

1 TIER 1 PRESENTATION & COMMAND CENTER LAYER

Streamlit UI - Port 8501

Streamlit

Incident Search Console
- Hybrid BM25 + Semantic
- Whole-dataset CSV Export

Search incidents... CSV

Multi-Agent AI Diagnosis Dashboard
- KPI meters
- RAG synthesis dossier
- Root cause cards

100K Dataset Explorer
- Filtered pagination
- over 100,000 cells

1–50 / 100000 ‹ ›

Benchmark & Evaluation Center
- Hit@k visualizers
- RAGAS metrics
- DeepEval reports

Human-in-the-Loop Feedback Portal
- Relevance ratings
- Audit logs

HTTP POST (JSON) ↓ | ↑ JSON Response (Render UI)

2 TIER 2 API GATEWAY & SECURITY INGRESS LAYER

FastAPI - Port 8000

FastAPI

Async REST Microservice

/search-incidents | /diagnose | /evaluate | /feedback

Ingress Guardrails

1. **Security & Anti-Injection Filter**
   - Regex XSS
   - Length bounds
   - <script> sanitization
2. **Telecom Domain Ontology Guardrail**
   - 70+ telecom keyword validation
   - Rejects OOD queries
   - HTTP 400 Bad Request

Out-of-Domain Query

Rejected Early (HTTP 400)

"best pizza near me"

Validated Query (In-Domain) ↓

3 TIER 3 INGESTION, CHUNKING & HYBRID RETRIEVAL ENGINE

Chunking Strategy

Structured Record Document Serialization (max_words=180, 27-column schema serialization preserving relational RF, KPI & alarm context into atomic 1-to-1 incident narratives)

Dual-Index Architecture

Sparse Lexical Index
Okapi BM25
- CountVectorizer
- dynamic IDF
- k1=1.5, b=0.75

Dense Vector Index
- Sentence-Transformers (all-MiniLM-L6-v2)
- 384-dimension embeddings
- Cosine Similiarity
- ChromaDB (persistent)

Reciprocal Score Fusion

Weighted Blend
0.65 Dense Semantic
+ **0.35** Sparse BM25

Metadata Pre-Filtering & Relaxation Engine
- State • City • Season
- Weather • Condition

(Nearest-match fallback)

Query → Search (Filtered Results)

Top-K Relevant Context ↓

4 TIER 4 MULTI-AGENT COGNITIVE ORCHESTRATION PIPELINE

10 Specialized Agents

1. **Incident Classifier Agent** — 8 telecom categories (latency, packet loss, handover, etc.)
2. **Prioritization Agent** — Composite severity scoring: CRITICAL / HIGH / MEDIUM / LOW
3. **Fault Risk Agent** — RandomForestClassifier 120 trees → Fault Risk %
4. **Performance Prediction Agent** — RandomForestRegressor 100 trees, depth 12 → Throughput Mbps
5. **Network Impact Assessor** — voice drops, retransmissions, QoE degradation
7. **Troubleshooting Recommender** — Deterministic SOP corrective actions
8. **Query Trend Analyzer** — Blocks temporal claims without dated evidence
9. **RAG Synthesizer** — Claude 3.5 / GPT-4o / Offline fallback
10. **LLM-as-Judge Evaluator** — 8-point audit (grounding, completeness, actionability)

Read / Write (Models, Vectors, Feedback) ↓↑

5 TIER 5 DATA PERSISTENCE & MODEL REPOSITORY

100,000 5G Cell Telemetry Dataset

SG_netops_data_100K.csv
- 27 telemetry columns: latency, packet loss, throughput, handover, alarms, ...

100,000 rows

Pretrained ML Artifacts (.joblib)
- throughput_predictor.joblib (RandomForestRegressor, 19 features)
- fault_risk_joblib (RandomForestClassifier, 20 features)

Storage & Feedback Store
- ChromaDB (storage/chroma)
- SQLite / JSON (audit trail)

ChromaDB | SQLite

6 TIER 6 AUTOMATED QUALITY & BENCHMARK HARNESS

Hit@k Keyword Retrieval Precision

100% Hit Rate ✓ 100%

RAGAS Generation Framework
- Faithfulness: 88%
- Response Relevancy: 93%
- Context Precision: 90%

88% / 93% / 90%

DeepEval Test Harness

100% Pass Rate across test suites ✓ 100%

End-to-End Data Flow: Tier 1 (UI) → Tier 2 (Guardrails) → Tier 3 (Hybrid Search) → Tier 4 (10-Agent Pipeline) → Tier 2 (JSON Response) → Tier 1 (Render UI)

Legend: Data Flow | Model Weights | Blocked Request